

Brendan J. Clark, Ph.D.

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EDUCATION

Ph.D. Atmospheric Science

January 2025

Rutgers University, New Brunswick

Advisors: Dr. Alan Robock, Dr. Lili Xia

Dissertation: "Stratospheric Sulfate Aerosol Climate Intervention Implications for Global Agriculture"

M.S. Atmospheric Science

November 2022

Rutgers University, New Brunswick

Advisors: Dr. Alan Robock, Dr. Lili Xia

Thesis: "Optimal climate intervention scenarios for crop production vary by nation"

B.S. Environmental Science

May 2020

University of Massachusetts, Amherst

Minors: GIS, Natural Resource Conservation

PROFESSIONAL EXPERIENCE

Postdoctoral Research Associate

March 2025-present

Cornell University

- Leveraging advanced statistical, machine learning, and process-based modeling to assess future impacts to tree growth and agriculture
- Collaborating with CarbonPlan doing statistical downscaling and cloud storage of climate model simulations

Graduate Research Assistant

August 2020-January 2025

Rutgers University

- Used the Community Earth System Model to assess impacts to global food production and nutritional quality under climate change and geoengineering
- Presented 15 first authors talks and posters
- Led cross-disciplinary research teams of up to 8 collaborators

PUBLICATIONS

PEER-REVIEWED ARTICLES (FIRST AUTHOR)

4. **B. Clark**, D. Visioni, S. Kothari, M. Lerdau. Land Models Likely Underestimate the Impact of Future Atmospheric Dryness on European Tree Growth. *Geophysical Research Letters* 53, e2026GL122759 **(2026)**. [10.1029/2026GL122759](https://doi.org/10.1029/2026GL122759)
3. **B. Clark**, A. Robock, L. Xia, S. S. Rabin, J. R. Guarin, J. Jaegermeyr. Stratospheric aerosol climate intervention could reduce crop nutritional value. *Environmental Research Letters*, 20, 114083 **(2025)**. [10.1088/1748-9326/ae1151](https://doi.org/10.1088/1748-9326/ae1151)

2. **B. Clark**, A. Robock, L. Xia, S. S. Rabin, J. R. Guarin, G. Hoogenboom, J. Jaegermeyr. Maize yield changes under sulfate aerosol climate intervention using three global gridded crop models. *Earth's Future* 13, e2024EF005269 (2025). [10.1029/2024EF005269](https://doi.org/10.1029/2024EF005269)
1. **B. Clark**, L. Xia, A. Robock, S. Tilmes, J. H. Richter, D. Vioni, S. S. Rabin. Optimal climate intervention scenarios for crop production vary by nation. *Nature Food* 4, 902–911 (2023). [10.1038/s43016-023-00853-3](https://doi.org/10.1038/s43016-023-00853-3)

PEER-REVIEWED ARTICLES (CO-AUTHOR)

2. N. Grant, A. Robock, L. Xia, J. Singh, **B. Clark**. Impacts on Indian agriculture due to stratospheric aerosol intervention using agroclimatic indices. *Earth's Future* 13, e2024EF005262 (2025). [10.1029/2024EF005262](https://doi.org/10.1029/2024EF005262)
1. F. Bowlick, **B. Clark**, et al. Understanding geocomputation education: A survey and syllabi informed review. *Research in Geographic Education* 23, 20-51 (2022).

FUNDING AND AWARDS

- | | |
|--|------|
| • CESM Workshop Travel Award (\$3,000) | 2026 |
| • Climate Intervention Network Travel Award (\$1,000) | 2025 |
| • NSF NCAR Exploratory Allocation Award (500,000 CPU hours) | 2025 |
| • GeoMIP Annual Meeting Travel Funding Award (\$3,500) | 2025 |
| • Rutgers Atmospheric Science Student Travel Support Award (\$1,500) | 2024 |
| • Rutgers Atmospheric Science Student Travel Support Award (\$1,200) | 2023 |
| • Rutgers Climate Institute Student Travel Support Award (\$1,000) | 2022 |
| • NSF NCAR University Small Allocation Award (200,000 CPU hours) | 2022 |
| • Governor's Award in Recognition of Environmental Stewardship | 2019 |
| • John and Abigail Adams Scholarship Award (\$7,000) | 2016 |
| • Stanley Z. Koplik Certificate of Mastery with Distinction Award | 2016 |

TEACHING EXPERIENCE

Guest Lecturer Spring 2026

Cornell University

Graduate course EAS 5555, Theory and Practice of Earth System Modeling

Guest Lecturer Spring 2026

Rutgers University

Undergraduate honors seminar 01:090:297, Climate Intervention: Where Science Meets Ethics

Teaching Assistant Fall 2019

University of Massachusetts

NRC 585, Introduction to GIS

Substitute Teacher 2018-2019

Algonquin Regional High School

CONFERENCE PRESENTATIONS (FIRST AUTHOR)

21. **B. Clark. Invited oral presentation**, June 2026. “Land models likely underestimate benefits to natural vegetation under stratospheric aerosol injection.” Gordon Research Conference, Newry, ME
20. **B Clark**, D. Vioni, S. Kothari, M. Lerda. **Oral presentation**, June 2026. “Land models likely underestimate the effect of atmospheric dryness on tree growth.” CESM Working Group Meeting, Boulder, CO
19. **B Clark**, D. Vioni, S. Kothari, M. Lerda. **Oral presentation**, March 2026. “Land models likely underestimate the effect of atmospheric dryness on European tree growth.” Coupled Model Intercomparison Project Community Workshop, Kyoto, Japan
18. **B Clark**, D. Vioni. **Poster**, March 2026. “Altered sunlight under stratospheric aerosol injection would have heterogenous impacts on vegetation.” Geoengineering Modeling Intercomparison Project annual meeting, Tokyo, Japan
17. **B Clark**, D. Vioni, S. Kothari, M. Lerda. **Oral presentation**, December 2025. “Land models likely underestimate the effect of atmospheric dryness on European tree growth.” American Geophysical Union meeting, New Orleans, LA
16. **B Clark**, D. Vioni, S. Kothari, M. Lerda. **Poster**, December 2025. “Using Dendrometer Data to Inform Projections of Carbon Storage under Solar adiation Modification.” American geophysical Union meeting, New Orleans, LA
15. **B. Clark. Poster**, May 2025 “Statistcally downscaled and bias adjusted ARISE-SAI and SSP2-4.5 scenarios.” DEGREES Global Forum, Cape Town, South Africa
14. **B. Clark**, A. Robock, L. Xia, S. Rabin, J Jaegermeyr, J. Guarin. **Oral presentation**, December 2024. “Stratospheric Aerosol Climate Intervention Could Reduce the Nutritional Value of Maize and Rice.” American Geophysical Union meeting, Washington D.C.
13. **B. Clark**, A. Robock, L. Xia, S. Rabin, J Jaegermeyr, J. Guarin. **Poster**, July 2024. “Sulfate Aerosol Climate Intervention Impacts on Maize Yield and Protein in Three Global Gridded Crop Models.” Geoengineering Modeling Intercomparison Project annual meeting, Cornell University
12. **B. Clark**, A. Robock, L. Xia, S. Rabin, J Jaegermeyr, J. Guarin. **Oral presentation**, February 2024. “Stratospheric Aerosol Climate Intervention Could Negatively Impact Crop Nutritional Quality.” Gordon Research Seminar, Newry, ME
11. **B. Clark**, A. Robock, L. Xia, S. Rabin, J Jaegermeyr, J. Guarin. **Poster**, February 2024. “Stratospheric Aerosol Climate Intervention Impacts on Crop Protein Content.” Gordon Research Conference, Newry, ME
10. **B. Clark**, A. Robock, L. Xia, S. Rabin, J Jaegermeyr, J. Guarin. **Oral Presentation**, December 2023. “Stratospheric Aerosol Climate Intervention Could Negatively Impact Crop Nutritional Quality.” American Geophysical Union meeting, San Francisco, CA
9. **B. Clark**, L. Xia, A. Robock, S. Tilmes, J. H. Richter, D. Vioni, S. S. Rabin. **Invited oral presentation**, July 2023. “The Optimal Climate Intervention Scenario for Crop Production Varies by Nation.” Exteter, UK

8. **B. Clark, A. Robock, L. Xia, S. Rabin, J. Singh, N. Grant. Poster**, July 2023. “Rutgers Impact Studies of Climate Intervention (RISCI) Laboratory Group Overview.” Geoengineering Modeling Intercomparison Project annual meeting, Exeter, UK
7. **B. Clark, A. Robock, L. Xia. Oral presentation**, June 2023. “A Proposal for a Multi-Crop Model Assessment of Stratospheric Aerosol Climate Intervention.” AgMIP-GGCM meeting, Columbia University
6. **B. Clark, A. Robock, L. Xia. Poster**, December 2022. “Can Crop Production be used as a Metric to Design Climate Intervention?” American Geophysical Union meeting, Chicago, IL
1. **B. Clark, A. Robock, L. Xia. Poster**, June 2022. “Impacts on Crop Production from Stratospheric Aerosol Climate Intervention: A Multi-Scenario Overview.” Gordon Research Conference, Newry, ME
4. **B. Clark, A. Robock, L. Xia. Invited oral presentation**, May 2022. “Crop Impacts from Stratospheric Aerosol Injection: A Multi-Scenario Overview.” University of Potsdam, Germany
2. **B. Clark, A. Robock, L. Xia. Oral presentation**, February 2022. “Discrepancies Between Fully Coupled and Offline CLM5 Crop Simulations.” NCAR Land Modeling Working Group Meeting, Boulder, CO
2. **B. Clark, A. Robock, L. Xia. Poster**, December 2021. “Impacts on Crop Production from Stratospheric Aerosol Injection” American Geophysical Union, New Orleans, LA
1. **B. Clark, A. Robock, L. Xia. Oral presentation**, October 2021. “Depicting Information and Remembering Your Audience: Impacts on Crop Production from Stratospheric Aerosol Climate Intervention” Climate Engineering in Context Conference, University of Potsdam, Germany

TECHNICAL SKILLS

- Python
 - Data analysis packages including xarray, numpy, pandas, scipy, and dask
 - Data visualization packages including matplotlib, seaborn, and plotly
 - Statistical modeling and machine learning packages including SciPy, statsmodels, scikit-learn, and seaborn
 - Jupyter notebooks and Google Colab
- Cloud computing, including AWS
- HTML, Linux, JavaScript, MATLAB, Fortran, R
- Version control and collaborative work using Git and GitHub
- Shell and command line – bash shell, batch scripting
- Parallelization using Python multiprocessing and Slurm scheduling
- High-performance computing environments including university clusters and NCAR’s Casper and Derecho

SERVICE

- Journal refereeing for: Earth's Future, Atmospheric Chemistry and Physics, Journal of Geophysical Research: Atmospheres, Environmental Research: Climate, PLOS Climate
- The Degrees Initiative grant proposal reviewer
- Reflective grant proposal reviewer
- Physicists Coalition for Nuclear Threat Reduction
- Science Policy and Advocacy at Rutgers

PROFESSIONAL AFFILIATIONS

- American Geophysical Union
- Ecological Society of America
- Agricultural Modeling Intercomparison Project
- Global Gridded Crop Modeling Intercomparison Project
- Geoengineering Modeling Intercomparison Project
- Climate Intervention Biology Working Group
- Inter-Sectoral Impact Model Intercomparison Project

SELECTED MEDIA COVERAGE

- "Climate Intervention Technologies May Create Winners and Losers in World Food Supply," *Rutgers Today*
- "Blocking the sun may fight climate change but kill our crops," *The Daily Beast*
- "Climate Tactics May Lower Crop Nutritional Value," *Mirage News*
- "Getting warmer: Slower forest growth means less carbon storage," *Cornell Chronicle*
- "What happens to our food supply when we dim the sun?" *Climate Chat Podcast*